

2-DAY SKILL CERTIFICATION PROGRAM

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Practical Assessments | Real Project Challenge | Portfolio Ready | Useful for Internships & Jobs

MainCrafts Technology
Real Skills. Real Projects. Real Opportunities.



DATA SCIENCE & DATA ANALYTICS USING PYTHON

RETAILPULSE 360

Sales, Profit & Customer Intelligence System

Difficulty: Intermediate

Duration: 2 Days / 48 Hours

Domain: Data Science & Data Analytics Using Python

Primary Language: Python

Environment: Google Colab

Dataset: Superstore / Global Superstore

Submission: Python Notebook + Report + Visual Evidence

MAINCRAFTS SKILLSPRINT

DATA SCIENCE & DATA ANALYTICS USING PYTHON

RETAILPULSE 360

Sales, Profit & Customer Intelligence System

48-HOUR PROJECT CHALLENGE

1. PROJECT BRIEF

A growing retail company has thousands of transactions but does not have a clear understanding of what is actually driving its business performance.

The management team wants answers to questions such as:

- Which products generate the most revenue?
- Which products generate the most profit?
- Which regions perform best?
- Which customer segments contribute the most?
- Are high-discount orders actually profitable?
- How are sales changing over time?
- Which products or regions require attention?
- What business actions can be taken from the data?

Your challenge is to build **RETAILPULSE 360**, a Python-based business analytics system that converts raw retail transaction data into meaningful business intelligence.

You are not being asked to simply create charts.

Your objective is to:

UNDERSTAND → CLEAN → ANALYZE → VISUALIZE → DISCOVER → EXPLAIN → RECOMMEND

At the end of the project, a business manager should be able to understand your findings without reading

your Python code.

2. YOUR MISSION

Build a complete analytical project that investigates:

SALES PERFORMANCE

Analyze:

- Total sales
- Sales by year
- Sales by month
- Sales by region
- Sales by category
- Sales by sub-category
- Sales by customer segment

PROFITABILITY

Analyze:

- Total profit
- Profit margin
- Profit by category
- Profit by region
- Profit by product
- Loss-making products
- Relationship between discount and profit

CUSTOMER ANALYTICS

Analyze:

- Number of customers
- Sales per customer
- Customer contribution
- High-value customers
- Customer segments

OPERATIONS

Analyze:

- Shipping modes
- Shipping duration
- Order priority
- Delivery performance

BUSINESS INSIGHTS

Finally, answer:

What should the company do based on the data?

3. THE BUSINESS SCENARIO

You have been hired as a **Junior Data Analyst** for a fictional retail company.

The management team has provided historical transaction data and asked you to prepare an analytical report.

The management does NOT want:

"Here are 10 graphs."

They want:

"What happened, why does it matter, and what should we pay attention to?"

Your final project should therefore contain both:

DATA ANALYSIS

and

BUSINESS INTERPRETATION.

4. PROJECT OBJECTIVES

By completing this project, you should demonstrate your ability to:

- work with a real-world tabular dataset
 - load data using Python
 - inspect a dataset
 - clean messy data
 - handle missing values
 - identify duplicates
 - convert dates
 - create calculated columns
 - perform exploratory data analysis
 - use Pandas for aggregation
 - use NumPy for calculations
 - create meaningful visualizations
 - perform time-based analysis
 - analyze customers and products
 - calculate KPIs
 - identify trends and anomalies
 - communicate business insights
 - make data-supported recommendations
-

5. TOOLS

Use free tools wherever possible.

Required

Python

Google Colab

Pandas

NumPy

Matplotlib

Seaborn

Optional

Plotly

Scikit-learn

GitHub

No paid software is required.

6. DATASET

For the official challenge, MainCrafts should provide students with the dataset directly.

A suitable public option is a **Superstore / Global Superstore** retail transaction dataset.

The dataset contains fields such as:

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Country
- Region
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Ship Mode

Public versions of the Global Superstore dataset are available on Kaggle and are commonly used for sales, profitability and business-analysis projects.

7. PROJECT STRUCTURE

Your final notebook should contain these sections:

01. Business Problem
02. Project Objective
03. Dataset Understanding
04. Environment & Libraries
05. Data Loading
06. Data Inspection
07. Data Cleaning

- 08. Feature Engineering
 - 09. KPI Analysis
 - 10. Sales Analysis
 - 11. Profitability Analysis
 - 12. Customer Analysis
 - 13. Product Analysis
 - 14. Regional Analysis
 - 15. Time-Series Analysis
 - 16. Discount & Profit Analysis
 - 17. Shipping Analysis
 - 18. Key Findings
 - 19. Business Recommendations
 - 20. Final Conclusion
 - 21. Optional Advanced Analysis
-

DAY 1 — DATA → CLEANING → EXPLORATION

STEP 1 — Create Your Notebook

Open Google Colab.

Create:

MainCrafts_SkillSprint_RetailPulse360_[YourName]

At the top:

RETAILPULSE 360

MainCrafts SkillSprint

Data Science & Data Analytics Using Python

Candidate:

Date:

Domain:

STEP 2 — Write the Business Problem

Before coding, write a short explanation.

Answer:

Question 1

What problem is the company trying to solve?

Question 2

Why is data analytics useful for a retail business?

Question 3

What decisions could management make using this data?

Question 4

What are the major KPIs you expect to investigate?

Write the answers in your own words.

STEP 3 — Import Libraries

Starter code:

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from datetime import datetime
```

```
print("Libraries loaded successfully.")
```

Your task

Explain what each library is being used for.

Do not simply paste the code without understanding it.

STEP 4 — Load the Dataset

Example:

```
df = pd.read_csv("superstore.csv")
```

```
print("Dataset loaded successfully.")
```

```
print("Rows:", df.shape[0])
```

```
print("Columns:", df.shape[1])
```

```
display(df.head())
```

If your supplied dataset is Excel:

```
df = pd.read_excel("superstore.xlsx")
```

STEP 5 — Understand the Dataset

Run:

```
df.info()
```

Then:

```
df.describe(include="all")
```

Then:

```
print(df.shape)
```

And:

```
display(df.head(10))
```

Identify:

- number of rows
- number of columns
- numerical columns
- categorical columns
- date columns
- identifier columns
- possible target/business metrics

Create a short **Data Dictionary** in your notebook.

Example:

Column	Meaning	Data Type
Order Date	Date order was placed	Date
Sales	Revenue from order	Numeric
Profit	Profit generated	Numeric
Quantity	Units ordered	Numeric
Discount	Discount applied	Numeric
Category	Product category	Categorical

STEP 6 — DATA QUALITY CHECK

Check missing values:

```
missing = df.isnull().sum()
```

```
display(  
    missing[missing > 0]  
)
```

Check duplicates:

```
print(  
    "Duplicate rows:",  
    df.duplicated().sum()  
)
```

Check data types:

```
print(df.dtypes)
```

Check unique values:

```
for column in df.select_dtypes(  
    include="object"  
)
```

```
print("\n", column)
print(df[column].nunique())
```

Your task

Create a short **Data Quality Summary**.

Example:

Total records:

Total columns:

Missing values:

Duplicate records:

Date fields:

Potential issues:

STEP 7 — CLEAN THE DATA

Convert dates:

```
df["Order Date"] = pd.to_datetime(
    df["Order Date"],
    errors="coerce"
)
```

```
df["Ship Date"] = pd.to_datetime(
    df["Ship Date"],
    errors="coerce"
)
```

Check:

```
print(df["Order Date"].dtype)
```

If duplicates are confirmed to be unnecessary:

```
df = df.drop_duplicates()
```

Do not delete columns or rows without explaining why.

STEP 8 — FEATURE ENGINEERING

Create useful analytical fields.

Year

```
df["Year"] = df["Order Date"].dt.year
```

Month

```
df["Month"] = df["Order Date"].dt.month
```

Month Name

```
df["Month_Name"] = df["Order Date"].dt.month_name()
```

Order Month

```
df["Order_Month"] = (  
    df["Order Date"]  
    .dt.to_period("M")  
    .astype(str)  
)
```

Shipping Days

```
df["Shipping_Days"] = (  
    df["Ship Date"] - df["Order Date"]  
)  
.dt.days
```

Profit Margin

```
df["Profit_Margin"] = (  
    df["Profit"] / df["Sales"]  
) * 100
```

Handle zero-sales cases appropriately if they exist.

STEP 9 — BUILD YOUR KPI PANEL

Calculate:

Total Sales

```
total_sales = df["Sales"].sum()
```

Total Profit

```
total_profit = df["Profit"].sum()
```

Total Orders

```
total_orders = df["Order ID"].nunique()
```

Total Customers

```
total_customers = df["Customer ID"].nunique()
```

Total Quantity

```
total_quantity = df["Quantity"].sum()
```

Average Order Value

```
average_order_value = (  
    total_sales / total_orders  
)
```

Overall Profit Margin

```
profit_margin = (  
    total_profit / total_sales  
) * 100
```

Display the KPIs clearly.

Example:

=====

=====

Total Sales:

Total Profit:

Total Orders:

Total Customers:

Units Sold:

Average Order Value:

Profit Margin:

=====

STEP 10 — SALES ANALYSIS

Question 1

Which category generates the most sales?

```
category_sales = (  
    df.groupby("Category")["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

display(category_sales)

Visualize it:

```
category_sales.plot(  
    kind="bar",  
    figsize=(8,5)  
)  
  
plt.title("Sales by Category")  
plt.xlabel("Category")  
plt.ylabel("Sales")  
plt.xticks(rotation=0)  
plt.show()
```

Your interpretation

Write:

The highest-sales category is _____.

It generated _____ in sales.

The lowest-sales category is _____.

This may indicate _____.

Do not invent explanations that aren't supported by the data.

STEP 11 — SALES BY REGION

```
region_sales = (  
    df.groupby("Region")["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

```
display(region_sales)
```

Create a visualization.

Then answer:

- Which region generates the most sales?
- Which region generates the least?
- What percentage of sales comes from the top region?
- Is high sales automatically associated with high profit?

That final question is important.

STEP 12 — PROFITABILITY ANALYSIS

Calculate profit by category:

```
category_profit = (  
    df.groupby("Category")["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

```
display(category_profit)
```

Then:

```
profit_region = (  
    df.groupby("Region")["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

```
display(profit_region)
```

Investigate:

Highest-sales category

vs.

Highest-profit category

They may not be the same.

If they differ, explain why this matters to management.

STEP 13 — LOSS-MAKING PRODUCTS

Find products with negative total profit.

```
product_profit = (  
    df.groupby("Product Name")["Profit"]  
    .sum()  
    .sort_values()  
)  
  
loss_products = product_profit[  
    product_profit < 0  
]  
  
display(  
    loss_products.head(10)  
)
```

Your task

Identify:

Top 10 loss-making products

and investigate:

- Sales
- Profit
- Discount
- Quantity
- Category
- Region

Don't simply say:

"These products are bad."

Explain what the data actually shows.

STEP 14 — DISCOUNT VS PROFIT

This is one of the most important business questions.

Create a scatter plot:

```
plt.figure(figsize=(9,6))  
  
sns.scatterplot(  
    data=df,  
    x="Discount",  
    y="Profit",  
    alpha=0.5  
)
```

```
plt.title(  
    "Relationship Between Discount and Profit"  
)  
  
plt.xlabel("Discount")  
plt.ylabel("Profit")
```

```
plt.show()
```

Then calculate:

```
discount_profit = (  
    df.groupby("Discount")["Profit"]  
    .mean()  
    .sort_index()  
)  
  
display(discount_profit)
```

Answer:

Does higher discount appear to be associated with lower profitability in this dataset?

Be careful:

Correlation or association does not automatically prove causation.

Your project will receive credit for recognizing this distinction.

STEP 15 — TIME-SERIES ANALYSIS

Aggregate monthly sales:

```
monthly_sales = (  
    df.groupby("Order_Month")["Sales"]  
    .sum()  
)
```

```
monthly_sales.plot(  
    figsize=(12,5)  
)
```

```
plt.title("Monthly Sales Trend")  
plt.xlabel("Month")  
plt.ylabel("Sales")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

Investigate:

- overall sales trend
- highest sales month
- lowest sales month
- year-over-year changes
- possible seasonal patterns

Calculate yearly sales:

```
yearly_sales = (  
    df.groupby("Year")["Sales"]  
    .sum()  
)  
  
display(yearly_sales)
```

STEP 16 — CUSTOMER ANALYTICS

Calculate customer-level sales:

```
customer_sales = (  
    df.groupby(  
        ["Customer ID", "Customer Name"]  
    )["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
)  
  
display(  
    customer_sales.head(10)  
)
```

Find top customers.

Then calculate:

```
customer_profit = (  
    df.groupby(  
        ["Customer ID", "Customer Name"]  
    )["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
)  
  
display(  
    customer_profit.head(10)
```

)

Important question

Are the customers generating the most sales also generating the most profit?

Investigate rather than assuming.

STEP 17 — CUSTOMER SEGMENT ANALYSIS

```
segment_analysis = (  
    df.groupby("Segment")  
    .agg(  
        Sales=("Sales", "sum"),  
        Profit=("Profit", "sum"),  
        Customers=("Customer ID", "nunique"),  
        Orders=("Order ID", "nunique")  
    )  
)
```

display(segment_analysis)

Create a visualization.

Then calculate:

```
segment_analysis["Profit_Margin"] = (  
    segment_analysis["Profit"]  
    / segment_analysis["Sales"]  
) * 100
```

display(segment_analysis)

Answer:

- Which segment has the highest sales?
 - Which has the highest profit?
 - Which has the highest profit margin?
 - Which segment has the largest customer base?
-

STEP 18 — SHIPPING ANALYSIS

Calculate average shipping time:

```
avg_shipping = (  
    df.groupby("Ship Mode")["Shipping_Days"]  
    .mean()  
    .sort_values()  
)
```

display(avg_shipping)

Investigate:

- average delivery time
- shipping mode usage
- relationship between shipping mode and profit
- regions with longer shipping times

Business question

Is faster shipping necessarily associated with higher profitability?

Use the dataset to investigate.

DAY 2 — DEEP ANALYSIS → INSIGHTS → BUSINESS REPORT

STEP 19 — FIND YOUR TOP 10 INSIGHTS

Now stop coding for a moment.

Review everything you've discovered.

Write **10 findings**.

But they must follow this structure:

Finding

What did the data show?

Evidence

What number/chart supports it?

Business Meaning

Why does it matter?

Example:

Finding:

Category X generated the highest sales.

Evidence:

Sales were ₹XXXX compared with ₹XXXX for Category Y.

Business Meaning:

The category represents a major revenue source and should be monitored closely for future sales planning.

STEP 20 — CREATE A BUSINESS QUESTION TABLE

Your final notebook should contain something like:

Business Question	Finding	Evidence	Action
Which category sells most?	...	...	...
Which category is most profitable?	...	...	...
Which region needs attention?	...	...	...
Does discount affect profit?	...	...	...

Who are the top customers?	...	...
----------------------------	-----	-----

This section is mandatory.

STEP 21 — BUILD A MINI ANALYTICS DASHBOARD

Create one final visual summary using Python.

It should contain:

KPI 1

Total Sales

KPI 2

Total Profit

KPI 3

Total Orders

KPI 4

Profit Margin

And at least:

- sales trend
- sales by category
- profit by region
- customer segment analysis
- discount/profit relationship

The dashboard can be created inside the notebook.

Optional: Students may build a Streamlit or Plotly-based interactive dashboard.

STEP 22 — OPTIONAL DATA SCIENCE EXTENSION

Students who finish the core project early can attempt a predictive analysis.

OPTION A — Sales Prediction

Use historical monthly data to create a simple forecasting/prediction experiment.

OPTION B — Profit Prediction

Use available transaction features to build a basic regression model.

OPTION C — Customer Segmentation

Use customer-level metrics such as:

- Recency
- Frequency
- Monetary value

to create customer groups.

This section is **optional**.

It should not be necessary for completing the basic 2-day challenge.

STEP 23 — FINAL BUSINESS RECOMMENDATIONS

You must provide **at least 5 recommendations**.

Each recommendation must be connected to evidence.

Example structure:

Recommendation 1

Observation:

...

Evidence:

...

Recommended Action:

...

Expected Business Purpose:

...

Do not write generic recommendations such as:

"The company should improve marketing."

Make them specific to your analysis.

STEP 24 — LIMITATIONS

A professional data analyst should also explain what the analysis cannot tell us.

Include at least **3 limitations**.

Possible examples:

- dataset represents historical transactions
- analysis does not prove causation
- external factors are not included
- customer satisfaction data may be unavailable
- marketing activity may not be represented
- future market conditions may differ

Students must identify limitations relevant to their own project.

STEP 25 — FINAL EXECUTIVE SUMMARY

At the end of the notebook, write a **one-page executive summary**.

It should answer:

What happened?

What matters?

What should management investigate?

What should management consider doing?

The executive summary should be understandable to a non-technical manager.

9. FINAL NOTEBOOK OUTPUT

Your completed notebook should approximately follow:

RETAILPULSE 360

|

|—— Executive Summary

|

|—— 01. Business Problem

|—— 02. Objectives

|—— 03. Dataset Overview

|—— 04. Data Dictionary

|—— 05. Data Quality

|—— 06. Data Cleaning

|—— 07. Feature Engineering

|—— 08. KPI Analysis

|—— 09. Sales Analysis

|—— 10. Profit Analysis

|—— 11. Product Analysis

|—— 12. Customer Analysis

|—— 13. Segment Analysis

|—— 14. Regional Analysis

|—— 15. Time-Series Analysis

|—— 16. Discount Analysis

|—— 17. Shipping Analysis

|—— 18. Business Questions

|—— 19. Top 10 Insights

|—— 20. Recommendations

|—— 21. Limitations

|—— 22. Final Conclusion

|—— 23. Optional Advanced Analysis

10. SUBMISSION FORMAT

Every participant must submit:

REQUIRED

1. Python Notebook

RetailPulse360_[YourName].ipynb

or a shareable Google Colab notebook.

2. Project Report

PDF containing:

- Project title

- Problem statement
- Objective
- Dataset
- Methodology
- Data cleaning
- Analysis
- Visualizations
- Findings
- Recommendations
- Limitations
- Conclusion

3. Evidence

Include screenshots or exported visualizations showing:

- data inspection
- cleaning
- KPI analysis
- major charts
- final findings
- final analytical output

4. README

If using GitHub:

RETAILPULSE-360/

|

|—— README.md

|—— RetailPulse360.ipynb

|—— dataset/

|—— visuals/

|—— report/

|—— requirements.txt

11. EVALUATION CRITERIA

The project will be evaluated out of 100 points.

Area	Points
Business Problem Understanding	10
Data Understanding & Cleaning	15
Feature Engineering	10
Exploratory Data Analysis	15
Statistical / KPI Analysis	10
Visualization Quality	10

Area	Points
Business Insights	15
Recommendations	5
Documentation & Presentation	10
TOTAL	100

Evaluation principle

The project is **not judged by how many charts are created**.

A student who creates 20 charts without meaningful interpretation should not automatically outperform a student who creates 8 strong analyses with clear business conclusions.

The evaluation should focus on:

Accuracy → Relevance → Reasoning → Interpretation → Communication

12. CERTIFICATE MAPPING

Program Invitation Letter

Issued after successful registration/acceptance.

Certificate of Participation

Issued after the student satisfies the defined participation requirements.

Project Completion Certificate

Issued after the required RetailPulse 360 project deliverables are successfully completed.

Project Title:

RETAILPULSE 360 — Sales, Profit & Customer Intelligence System

Certificate of Achievement

Issued according to the published evaluation criteria and achievement threshold.

13. 48-HOUR PLAN

DAY 1

Hours 1–2

Understand business problem + dataset

Hours 2–4

Data inspection + data cleaning

Hours 4–6

Feature engineering + KPIs

Hours 6–8

Sales + profitability analysis

Hours 8–10

Customer + product analysis

Hours 10–12

Regional + time-series analysis

DAY 2

Hours 1–3

Discount + shipping analysis

Hours 3–5

Deep investigation + business questions

Hours 5–7

Top 10 insights

Hours 7–8

Recommendations

Hours 8–10

Mini analytics dashboard

Hours 10–11

Documentation

Hours 11–12

Final review + submission

14. FINAL PROJECT STANDARD

A successful participant should be able to explain:

"I took raw retail transaction data, cleaned and transformed it using Python, analyzed sales, profitability, customers, products, regions and time trends, created visualizations, identified business patterns, and converted those findings into data-supported recommendations."

That is the skill this challenge is designed to demonstrate.

15. OPTIONAL ADVANCED PORTFOLIO EXTENSION

Participants who want to go beyond the minimum can add:

Interactive Dashboard

Using:

- Streamlit
- Plotly

Predictive Analysis

Using:

- Scikit-learn

Customer Segmentation

Using:

- RFM analysis
- K-Means clustering

Automated Report

Generate a reusable analytical report from the dataset.

These extensions are **portfolio enhancements**, not mandatory requirements for the basic 2-day challenge.

Official/free resources for the students

I would give participants a **resource pack** rather than making them search Google themselves.

- **Google Colab** — free, browser-based Jupyter environment with no setup; Google specifically describes it as suitable for data science and education. [Google Colab](#)
- **Pandas documentation** — the core Python library for working with tabular data, DataFrames, cleaning and analysis. [Pandas documentation](#)
- **Pandas getting-started tutorials** — useful for students who are learning Pandas during the challenge. [Pandas Getting Started](#)
- **Seaborn documentation** — useful for statistical/data visualization; Seaborn is built around Matplotlib and works closely with Pandas. [Seaborn](#)
- **Matplotlib documentation** — for standard Python charts and visualizations. [Matplotlib](#)
- **NumPy documentation** — for numerical computing and array-based operations. [NumPy](#)
- **Kaggle Superstore dataset** — one current public version contains retail transaction fields such as order dates, customer segments, categories, sales, discounts and profit. [Global Superstore Dataset](#)
- **Alternative Superstore dataset** — another current public version explicitly describes the dataset as suitable for EDA, business analytics and ML modeling. [Superstore Dataset](#)